INTRODUCTORY BUSINESS STATISTICS

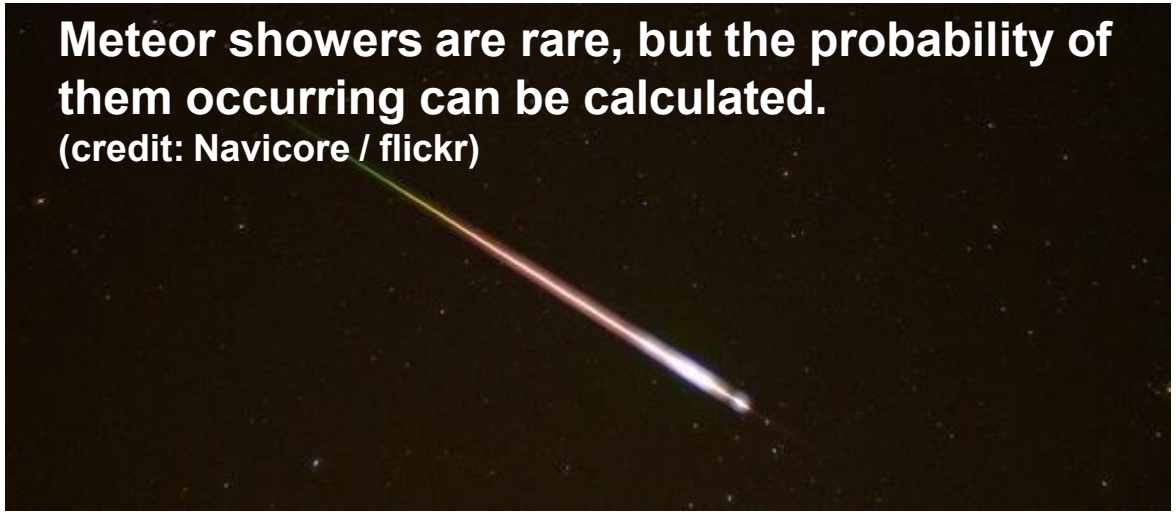
Chapter 3 PROBABILITY TOPICS



Chapter Objectives

By the end of this chapter, the student should be able to:

- Understand and use the terminology of probability.
- Determine whether two events are mutually exclusive and whether two events are independent.
- Calculate probabilities using the Addition Rules and Multiplication Rules.
- Construct and interpret Contingency Tables.
- Construct and interpret Probability Trees/Tree Diagrams.
- Construct and interpret Venn Diagrams.



Meteor showers are rare, but the probability of them occurring can be calculated.
(credit: Navicore / flickr)

Probability and Chance Experiment

Probability is a measure that is associated with how certain we are of outcomes of a particular experiment or activity.

An experiment is a planned operation carried out under controlled conditions.

If the result is not predetermined, then the experiment is said to be a chance experiment.

Flipping one fair coin twice is an example of a chance experiment.



Sample Space and Outcome

A result of an experiment is called an outcome. The sample space of an experiment is the set of all possible outcomes.

Three ways to represent a sample space are:

- to list the possible outcomes
- to create a tree diagram
- to create a Venn diagram

The uppercase letter S is used to denote the sample space.

For example, if you flip one fair coin, S = {H, T} where H = heads and T = tails are the outcomes.

Events and Probability of Events

An event is any combination of outcomes. Upper case letters like A and B represent events.

To flip one fair coin, event A might be getting at most one head. The probability of an event A is written *P(A).*

The probability of any outcome is the long-term relative frequency of that outcome. Probabilities are between zero and one, inclusive (that is, zero and one and all numbers between these values).

P(A) = 0 means the event A can never happen.

P(A) = 1 means the event A always happens.

P(A) = 0.5 means the event A is equally likely to occur or not to occur.

Events and Probability of Events

For example, if you flip one fair coin repeatedly (from 20 to 2,000 to 20,000 times) the relative frequency of heads approaches 0.5 (the probability of heads).

Equally likely means that each outcome of an experiment occurs with equal probability.

For example, if you toss a fair, six-sided die, each face (1, 2, 3, 4, 5, or 6) is as likely to occur as any other face. If you toss a fair coin, a Head (H) and a Tail (T) are equally likely to occur. If you randomly guess the answer to a true/false question on an exam, you are equally likely to select a correct answer or an incorrect answer.

Events and Probability of Events

For example, if you toss a fair dime and a fair nickel, the sample space is {HH, TH, HT, TT} where T = tails and H = heads. The sample space has four outcomes. A = getting one head. There are two outcomes that meet this condition, A = {HT, TH}, since the outcomes HH, TH, HT, TT are equally likely, we have:

$$P(A) = \frac{\text{Number of outcomes in event A}}{\text{Number of outcomes in sample space}} = \frac{2}{4} = 0.5$$

Events and Probability of Events

Example: you roll one fair six-sided die, with the numbers {1, 2, 3, 4, 5, 6} on its faces. Let event E = rolling a number that is at least five. There are two outcomes {5, 6}.

$$p(E) = \frac{2}{6}$$

If you were to roll the die only a few times, you would not be surprised if your observed results did not match the probability. If you were to roll the die a very large number of times, you would expect that, overall, 2/6 of the rolls would result in an outcome of “at least five”. You would not expect exactly 2/6.

The long-term relative frequency of obtaining this result would approach the theoretical probability of 2/6 as the number of repetitions grows larger and larger.

Events and Probability of Events

Example: Use the following information to answer the next six questions. A jar of 150 jelly beans contains 22 red jelly beans, 38 yellow, 20 green, 28 purple, 26 blue, and the rest are orange.

Let B = the event of getting a blue jelly bean. P(B)=? $p(B) = \frac{26}{150}$

Let G = the event of getting a green jelly bean. P(G)=? $p(G) = \frac{20}{150}$

Let O = the event of getting an orange jelly bean. P(O)=? $p(O) = \frac{16}{150}$

Let P = the event of getting a purple jelly bean. P(p)=? $p(p) = \frac{28}{150}$

Let R = the event of getting a red jelly bean. P(R)=? $p(R) = \frac{22}{150}$

Let Y = the event of getting a yellow jelly bean. P(Y)=? $p(Y) = \frac{38}{150}$

A OR B Event, A AND B Event

An outcome is in the event A OR B if the outcome is in A or is in B or is in both A and B.

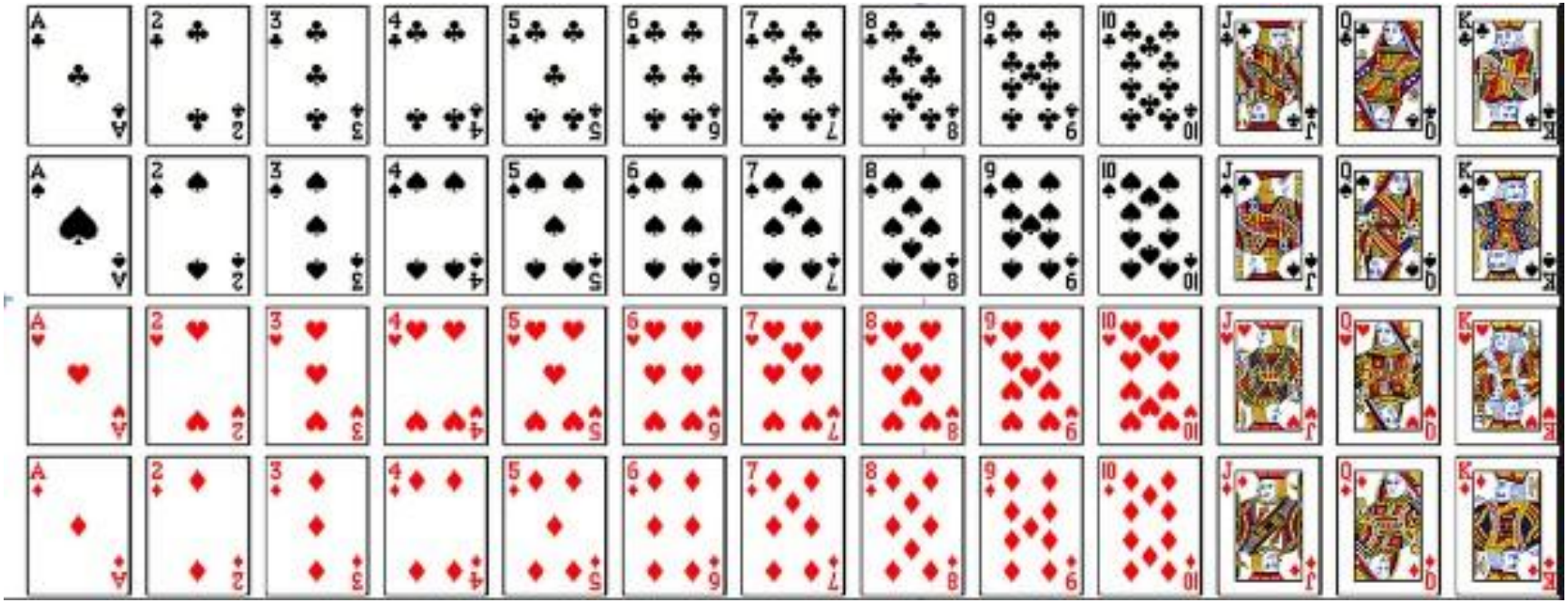
For example, let A = {1, 2, 3, 4, 5} and B = {4, 5, 6, 7, 8}.
A OR B = {1, 2, 3, 4, 5, 6, 7, 8}.

Notice that 4 and 5 are NOT listed twice.

An outcome is in the event A AND B if the outcome is in both A and B at the same time.

For example, let A and B be {1, 2, 3, 4, 5} and {4, 5, 6, 7, 8}, respectively. Then

A AND B = {4, 5}.



The complement of event A, A′

The complement of event A is denoted A′ (read “A prime”).

A′ consists of all outcomes that are NOT in A.

Notice that $P(A) + P(A') = 1$.

For example, let S = {1, 2, 3, 4, 5, 6} and let A = {1, 2, 3, 4}. Then,

A′ = {5, 6}.

$P(A) = 4/6$, $P(A') = 2/6$, and

$P(A) + P(A') = 4/6 + 2/6 = 1$.

The conditional probability

The conditional probability of A given B is written P(A|B).

P(A|B) is the probability that event A will occur given that the event B has already occurred.

A conditional reduces the sample space. We calculate the probability of A from the reduced sample space B.

The conditional probability

Example: Suppose we toss one fair, six-sided die. The sample space S = {1, 2, 3, 4, 5, 6}

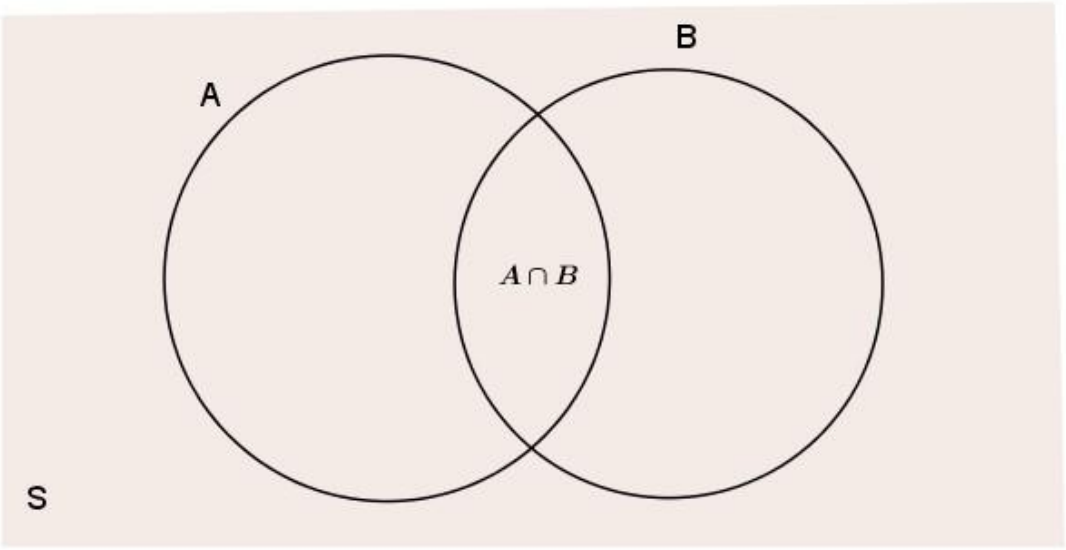
Let A = face is 2 or 3 and B = face is even (2, 4, 6). To calculate P(A|B), we count the number of outcomes 2 or 3 in the reduced sample space B = {2, 4, 6}. Then we divide that by the number of outcomes B (rather than S).

P(A|B) = 1/3.

3.2
INDEPENDENT AND MUTUALLY EXCLUSIVE EVENTS

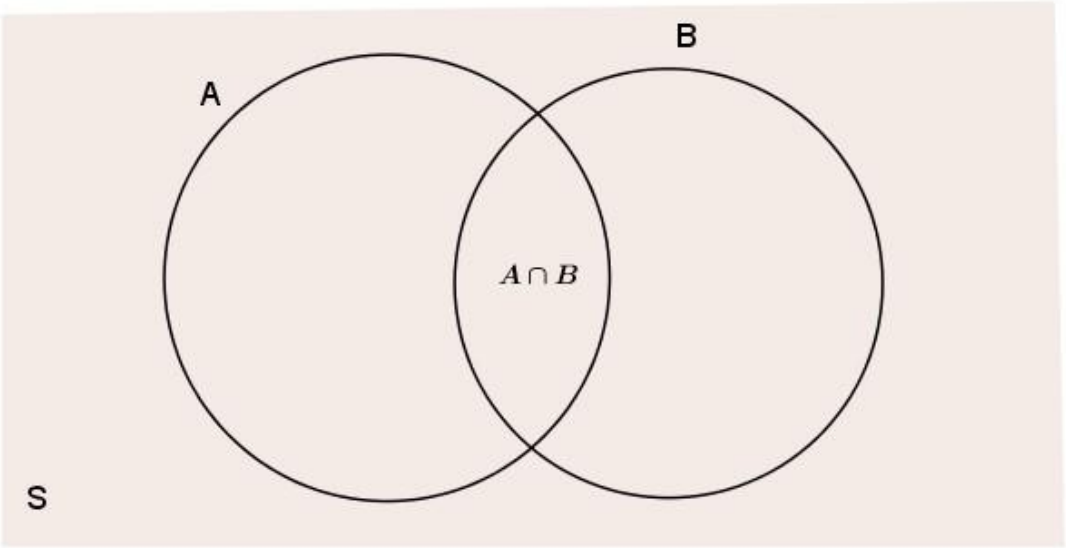
The conditional probability Again

The Venn Diagram below is a way to represents the basic Idea of probability such as sample space, an event, A AND B, A OR B, and conditional probability. The rectangle represents the sample space, and the area of it represents the probability to get the events in the sample space which is 100% or 1.



The conditional probability Again

The circles represent events, and areas of the circles represent probability of the events. The area they share represents p(A and B), and the total enclosed by the circles represents p(A OR B). If we have equally likely outcomes in the rectangle or circles, then to find the probabilities we count the outcomes.



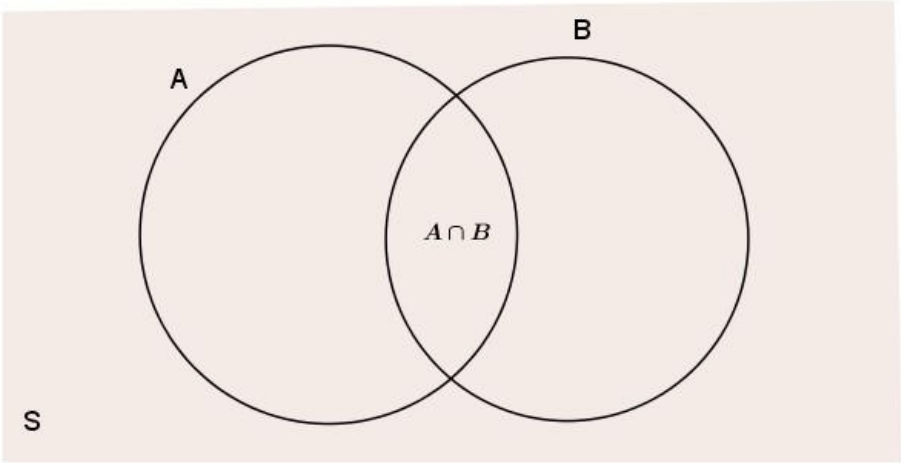
The conditional probability Again

The formula to find conditional probability can be made more intuitive by the use of this Venn diagram.

$$p(B \mid A) = \frac{p(A \cap B)}{p(A)}$$

Note that the p(B|A) is the area of B that is in A, or “how much of B given that we are in A?”

And, notice that if we are in circle A, then the part of B that is in A is how much A and B have in common! Thus, the numerator of the formula.



The conditional probability Again

$$p(B \mid A) = \frac{p(A \cap B)}{p(A)}$$

This formula also gives us a formula to calculate the probability of A and B if we solve for it in the above formula:

$$p(B \mid A) = \frac{p(A \cap B)}{p(A)} \rightarrow p(A \cap B) = p(A)\,p(B \mid A)$$

Independent Events

Two events A and B are independent if the knowledge that one occurred does not affect the chance the other occurs. Two events are independent if the following are true:

$P(A|B) = P(A)$

$P(B|A) = P(B)$

$P(A \text{ AND } B) = P(A)P(B)$

For example, the outcomes of two roles of a fair die are independent events. The outcome of the first roll does not change the probability for the outcome of the second roll.

To show two events are independent, you must show only one of the above conditions. If two events are NOT independent, then we say that they are dependent.

Sampling With Replacement and Without Replacement

With replacement:

If each member of a population is replaced after it is picked, then that member has the possibility of being chosen more than once. When sampling is done with replacement, then events are considered to be independent, meaning the result of the first pick will not change the probabilities for the second pick.

Without replacement:

When sampling is done without replacement, each member of a population may be chosen only once. In this case, the probabilities for the second pick are affected by the result of the first pick. The events are considered to be dependent or not independent.

Sampling with replacement example:

Suppose you pick three cards with replacement. The first card you pick out of the 52 cards is the Q of spades. You put this card back, reshuffle the cards and pick a second card from the 52-card deck. It is the ten of clubs. You put this card back, reshuffle the cards and pick a third card from the 52-card deck. This time, the card is the Q of spades again. Your picks are {Q of spades, ten of clubs, Q of spades}. You have picked the Q of spades twice. You pick each card from the 52-card deck.

Sampling without replacement example:

Suppose you pick three cards without replacement. The first card you pick out of the 52 cards is the K of hearts. You put this card aside and pick the second card from the 51 cards remaining in the deck. It is the three of diamonds. You put this card aside and pick the third card from the remaining 50 cards in the deck. The third card is the J of spades. Your picks are {K of hearts, three of diamonds, J of spades}. Because you have picked the cards without replacement, you cannot pick the same card twice.

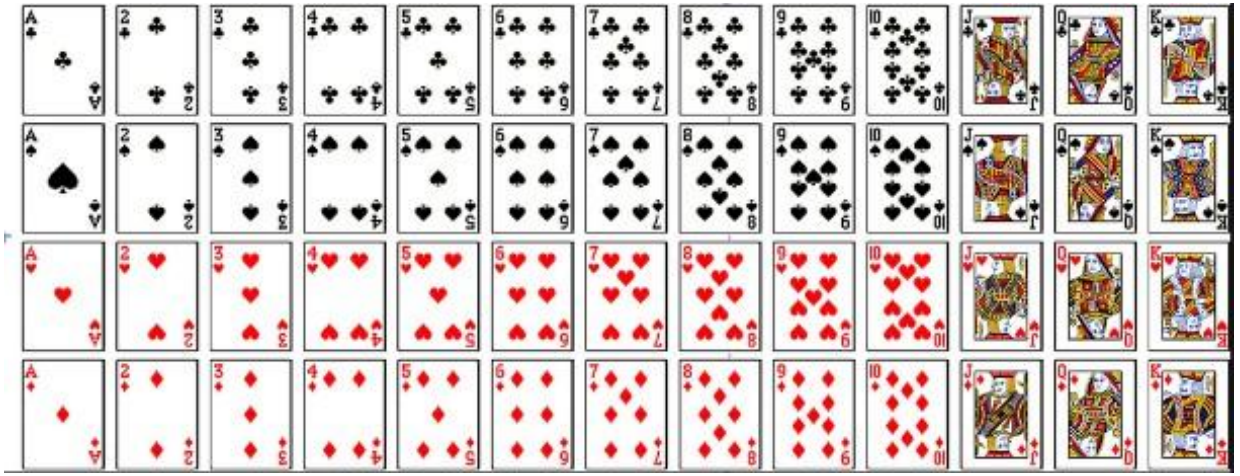
Example

You have a fair, well-shuffled deck of 52 cards. Suppose that you sample four cards without replacement. Which of the following outcomes are possible? Answer the same question for sampling with replacement.

QS, 1D, 1C, QD

KH, 7D, 6D, KH

QS, 7D, 6D, KS



without replacement: 1. Possible; 2. Impossible, 3. Possible

with replacement: 1. Possible; 2. Possible, 3. Possible

Mutually Exclusive Events

A and B are mutually exclusive events if they cannot occur at the same time. This means that A and B do not share any outcomes and P(A AND B) = 0.

For example

Suppose the sample space S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Let A = {1, 2, 3, 4, 5}, B = {4, 5, 6, 7, 8}, and C = {7, 9}.

A AND B = {4, 5}. P(A AND B) = 2/10 and is not equal to zero.

Therefore, A and B are not mutually exclusive. A and C do not have any numbers in common so P(A AND C) = 0. Therefore, A and C are mutually exclusive.

If it is not known whether A and B are mutually exclusive, assume they are not until you can show otherwise.

Example

Flip two fair coins. (This is an experiment.)

The sample space is {HH, HT, TH, TT}. The outcomes are HH, HT, TH, and TT and they are equally likely.

Let A = the event of getting at most one tail. Then A can be written as {HH, HT, TH}. The outcome HH shows zero tails. HT and TH each show one tail.

Let B = the event of getting all tails. B can be written as {TT}.

B is the complement of A, so B = A'

Also, P(A) + P(B) = P(A) + P(A') = 1.

The probabilities for A and for B are P(A) = 3/4 and P(B) = 1/4.

Example

Flip two fair coins. (This is an experiment.)

The sample space is {HH, HT, TH, TT}. The outcomes are HH, HT, TH, and TT and they are equally likely.

Let C = the event of getting all heads. C = {HH}.
Since B = {TT}, P(B AND C) = 0.

B and C are mutually exclusive.

Let D = event of getting more than one tail. D = {TT}. P(D) = 1/4
Let E = event of getting a head on the first roll. (This implies you can get either a head or tail on the second roll.) E = {HT, HH}. P(E) = 2/4

Find the probability of getting at least one (one or two) tail in two flips. Let F = event of getting at least one tail in two flips. F = {HT, TH, TT}. P(F) = 3/4

3.3
TWO BASIC RULES OF PROBABILITY

The Multiplication Rule

If A and B are two events defined on a sample space, then:

P(A AND B) = P(B)P(A|B).

$$p(A \cap B) = p(A)\,p(B \mid A) \qquad or \qquad p(B \mid A) = \frac{p(A \cap B)}{p(A)}$$

If A and B are independent, then $p(B \mid A) = p(B)$, *and*

$$p(A \cap B) = p(A)\,p(B)$$

The Addition Rule

If A and B are defined on a sample space, then:

P(A OR B) = P(A) + P(B) - P(A AND B).

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

If A and B are mutually exclusive, then $p(A \cap B) = 0$, *and*

$$p(A \cup B) = p(A) + p(B)$$

Example

Juan is trying to choose where to go on vacation. His two choices are: A = New Zealand and B = Alaska

Juan can only afford one vacation. The probability that he chooses A is P(A) = 0.6 and the probability that he chooses B is P(B) = 0.35.

Note that:

P(A AND B) = 0 because Juan can only afford to take one vacation.

P(A OR B) = P(A) + P(B) = 0.6 + 0.35 – 0 = 0.95.

The probability that he does not choose to go anywhere on vacation must be 1– 0.95 = 0.05.

Example

A student goes to the library. Let events

B = the student checks out a book and

D = the student checks out a DVD.

Suppose that P(B) = 0.40, P(D) = 0.30 and P(D|B) = 0.5.

Find P(B AND D).

Find P(B OR D).

$$p(B \cap D) = p(B)\,p(D \mid B)$$

$$p(B \cap D) = (0.40)(0.5) = 0.20.$$

$$p(B \cup D) = p(B) + p(D) - p(B \cap D)$$

$$p(B \cup D) = 0.40 + 0.30 - 0.20 = 0.50.$$

Example

Studies show that about one woman in seven (approximately 14.3%) who live to be 90 will develop breast cancer. Suppose that of those women who develop breast cancer, a test is negative 2% of the time. Also suppose that in the general population of women, the test for breast cancer is negative about 85% of the time. Let B = woman develops breast cancer and let N = tests negative. Suppose one woman is selected at random.

a. What is the probability that the woman develops breast cancer? What is the probability that woman tests negative?

a. $P(B) = 0.143$; $P(N) = 0.85$

b. Given that the woman has breast cancer, what is the probability that she tests negative?

b. $P(N|B) = 0.02$

c. What is the probability that the woman has breast cancer AND tests negative?

c. $P(B \text{ AND } N) = P(B)P(N|B) = (0.143)(0.02) = 0.0029$

Example

Studies show that about one woman in seven (approximately 14.3%) who live to be 90 will develop breast cancer. Suppose that of those women who develop breast cancer, a test is negative 2% of the time. Also suppose that in the general population of women, the test for breast cancer is negative about 85% of the time. Let B = woman develops breast cancer and let N = tests negative. Suppose one woman is selected at random.

d. What is the probability that the woman has breast cancer or tests negative?

d. $P(B \text{ OR } N) = P(B) + P(N) - P(B \text{ AND } N) = 0.143 + 0.85 - 0.0029 = 0.9901$

e. Are having breast cancer and testing negative independent events?

e. No. P(N) = 0.85; P(N|B) = 0.02. So, P(N|B) does not equal P(N).

f. Are having breast cancer and testing negative mutually exclusive?

f. No. P(B AND N) = 0.0029. For B and N to be mutually exclusive, P(B AND N) must be zero.

3.4
CONTINGENCY TABLES

Contingency Table

A contingency table provides a way of portraying data that can facilitate calculating probabilities. The table helps in determining conditional probabilities quite easily. The table displays sample values in relation to two different variables that may be dependent or contingent on one another.

Example:
Suppose a study of speeding violations and drivers' use of cell phones produced the following fictional data:

	Speeding violation in the last year	No speeding violation in the last year	Total
Uses cell phone while driving	25	280	305
Does not use cell phone while driving	45	405	450
Total	70	685	755

The total number of people in the sample is 755. The row totals are 305 and 450. The column totals are 70 and 685.
Notice that 305 + 450 = 755 and 70 + 685 = 755.

Contingency Table

	Speeding violation in the last year	No speeding violation in the last year	Total
Uses cell phone while driving	25	280	305
Does not use cell phone while driving	45	405	450
Total	70	685	755

a. Find *P*(Driver is a cell phone user).

$$\text{a. } \frac{\text{number of cell phone users}}{\text{total number in study}} = \frac{305}{755}$$

b. Find P(driver had no violation in the last year).

$$\text{b. } \frac{\text{number that had no violation}}{\text{total number in study}} = \frac{685}{755}$$

c. Find *P*(Driver had no violation in the last year AND was a cell phone user).

$$\text{c. } \frac{280}{755}$$

d. Find *P*(Driver is a cell phone user OR driver had no violation in the last year).

$$\text{d. } \left(\frac{305}{755} + \frac{685}{755}\right) - \frac{280}{755} = \frac{710}{755}$$

Contingency Table

	Speeding violation in the last year	No speeding violation in the last year	Total
Uses cell phone while driving	25	280	305
Does not use cell phone while driving	45	405	450
Total	70	685	755

e. Find *P*(Driver is a cell phone user GIVEN driver had a violation in the last year).

e. $\frac{25}{70}$ (The sample space is reduced to the number of drivers who had a violation.)

f. Find *P*(Driver had no violation last year GIVEN driver was not a cell phone user)

f. $\frac{405}{450}$ (The sample space is reduced to the number of drivers who were not cell phone users.)

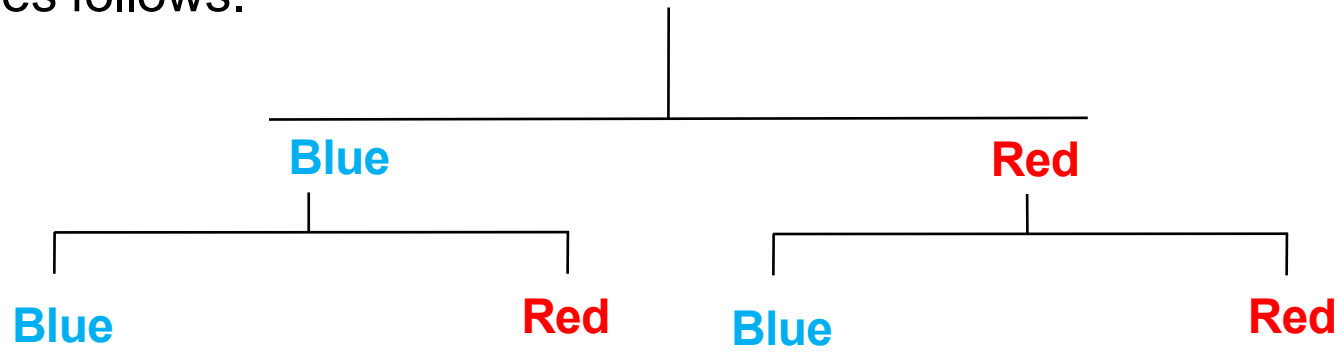
3.5
TREE AND VENN DIAGRAMS

Tree Diagram

A tree diagram is a special type of graph used to determine the outcomes of an experiment. It consists of “branches” that are labeled with either frequencies or probabilities. Tree diagrams can make some probability problems easier to visualize and solve. The following examples illustrate how to use a tree diagram.

Example

In an urn, there are 2 balls. 1 red ball (R) and 1 blue ball (B). Draw two balls, one at a time, with replacement. "With replacement" means that you put the first ball back in the urn before you select the second ball. The tree diagram using frequencies that show all the possible outcomes follows.

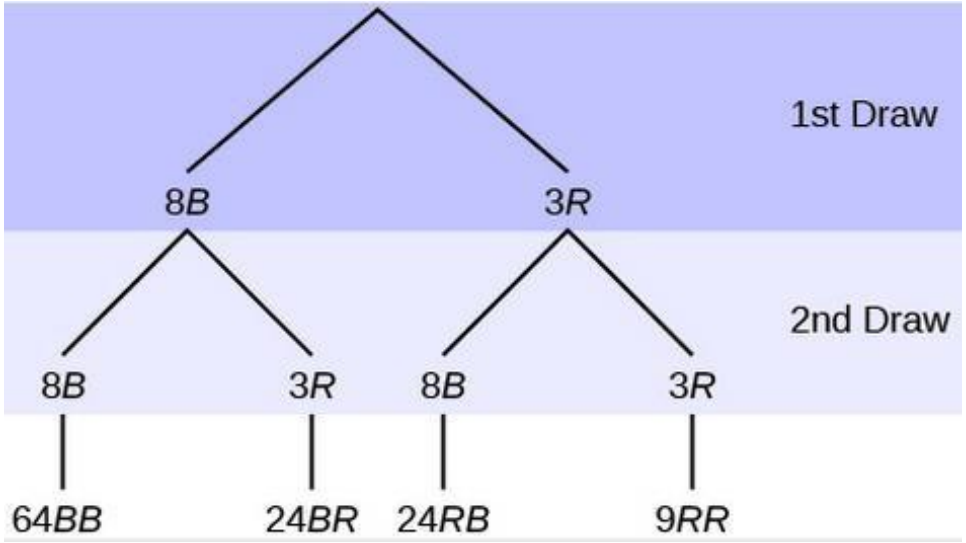


So, there are 4 outcomes in the sample space: **BB**, **BR**, **RB**, **RR** .

Tree Diagram

Example

It would be too many branches to deal with if there are more balls, so we simplify the process by having a branch represent multiple branches. In an urn, there are 11 balls. Three balls are red (R) and eight balls are blue (B). Draw two balls, one at a time, with replacement. “With replacement” means that you put the first ball back in the urn before you select the second ball. The tree diagram using frequencies that show all the possible outcomes follows.



The first set of branches represents the first draw. The second set of branches represents the second draw. Each of the outcomes is distinct. In fact, we can list each red ball as R1, R2, and R3 and each blue ball as B1, B2, B3, B4, B5, B6, B7, and B8. Then, we have:

3R: R1 R2 R3 , actually 3 branches represented by 3R

8B: B1 B2 B3 B4 B5 B6 B7 B8, actually 8 branches

9RR: R1R1 R1R2 R1R3 R2R1 R2R2 R2R3 R3R1 R3R2 R3R3, 9 branches, we get this by multiplying 3 x 3.

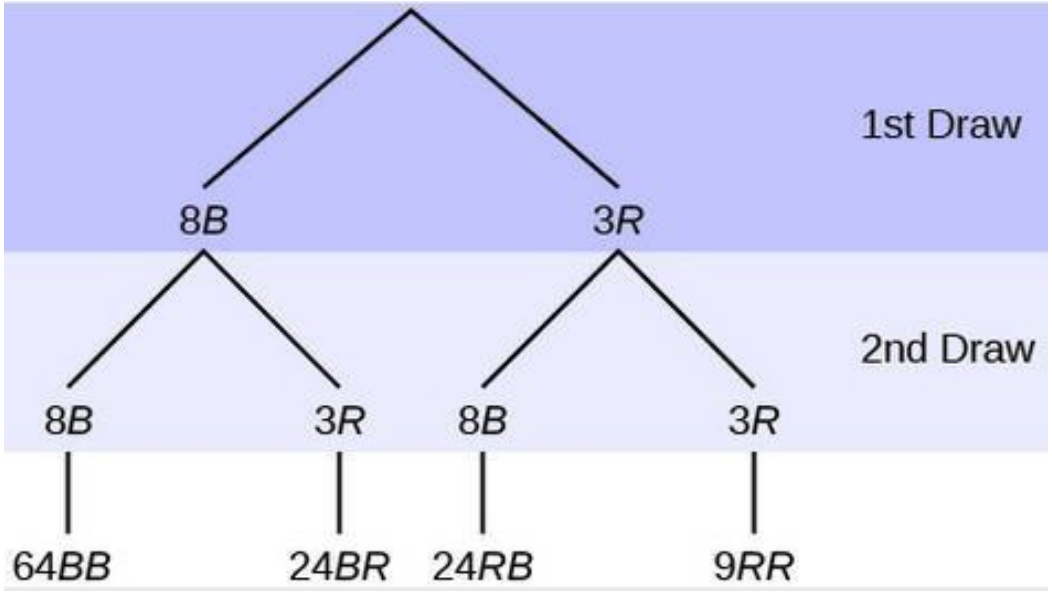
24RB: 24 branches, 3 x 8.

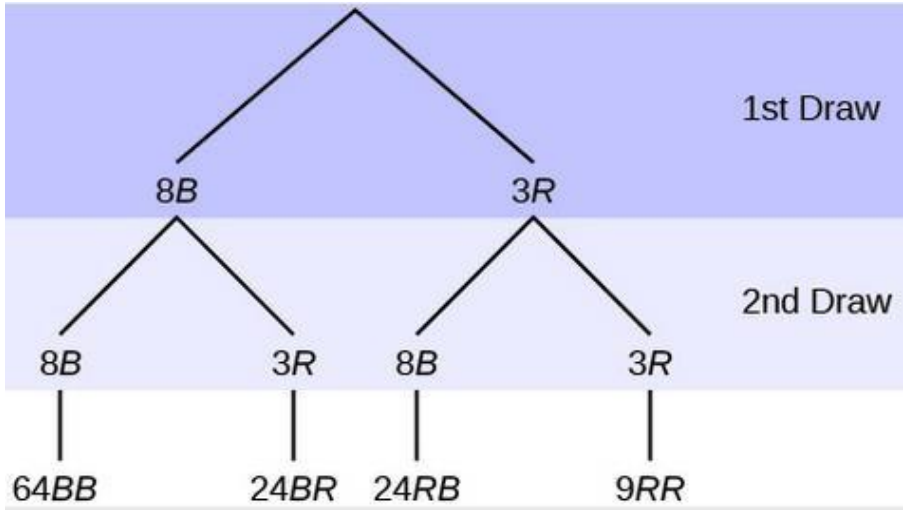
24BR: 24 branches, 8 x 3.

64BB: 64 branches, 8 x 8.

There are two position to fill with replacement, so there are 11(11) = 121 outcomes, the size of the sample space. We could also add the ends of the branches.

64 + 24 + 24 + 9 = 121





a. List the 24 BR outcomes: B1R1, B1R2, B1R3, ...

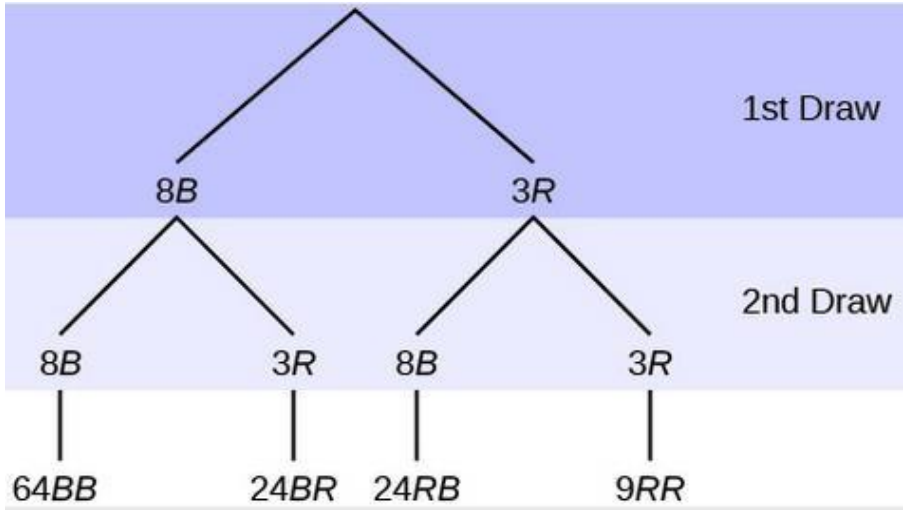
b. B1R1 B1R2 B1R3 B2R1 B2R2 B2R3 B3R1 B3R2 B3R3 B4R1 B4R2 B4R3 B5R1 B5R2 B5R3 B6R1 B6R2 B6R3 B7R1 B7R2 B7R3 B8R1 B8R2 B8R3

c. Using the tree diagram, calculate P(RR).

b. P(RR) = (3/11)(3/11) = 9/121

c. Using the tree diagram, calculate P(RB OR BR).

c. P(RB OR BR) = (3/11)(8/11) + (8/11)(3/11) = 48/121



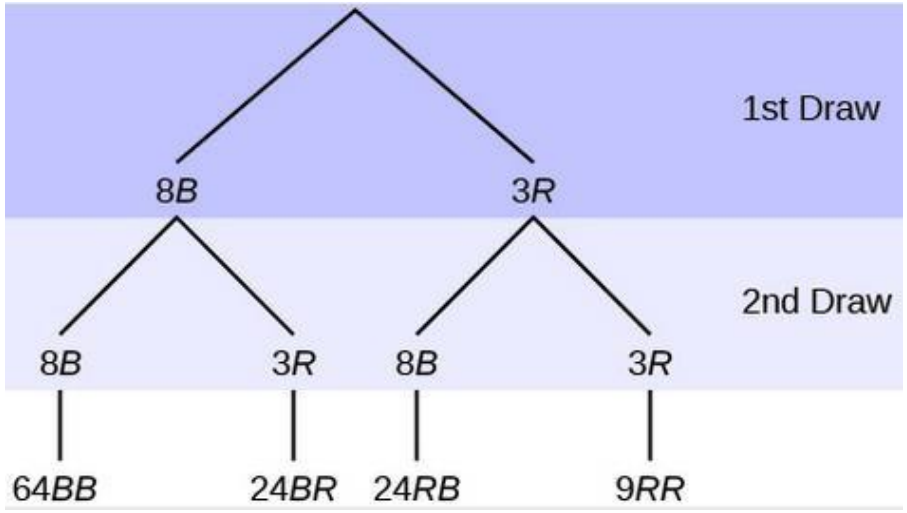
d. Using the tree diagram, calculate P(R on 1st draw AND B on 2nd draw).

d. P(R on 1st draw AND B on 2nd draw) = P(RB) = (3/11)(8/11) = 24/121

e. Using the tree diagram, calculate P(R on 2nd draw GIVEN B on 1st draw).

e. P(R on 2nd draw GIVEN B on 1st draw) = P(R on 2nd|B on 1st) = 24/88 = 3/11

This problem is a conditional one. The sample space has been reduced to those outcomes that already have a blue on the first draw. There are 24 + 64 = 88 possible outcomes (24 BR and 64 BB). Twenty-four of the 88 possible outcomes are BR. 24/88 = 3/11.



f. Using the tree diagram, calculate P(BB).

f. P(BB) = 64/121=(8/11)(8/11)

g. Using the tree diagram, calculate P(B on the 2nd draw given R on the first draw).

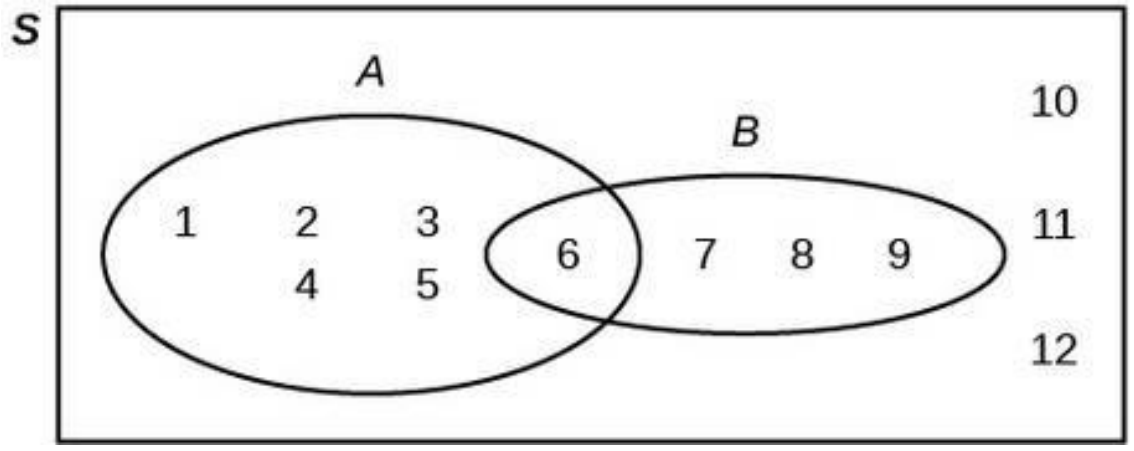
g. P(B on 2nd draw|R on 1st draw) = 8/11

There are 9 + 24 outcomes that have R on the first draw (9 RR and 24 RB). The sample space is then 9 + 24 = 33. 24 of the 33 outcomes have B on the second draw. The probability is then 24/33.

Venn Diagram

A Venn diagram is a picture that represents the outcomes of an experiment. It generally consists of a box that represents the sample space S together with circles or ovals. The circles or ovals represent events.

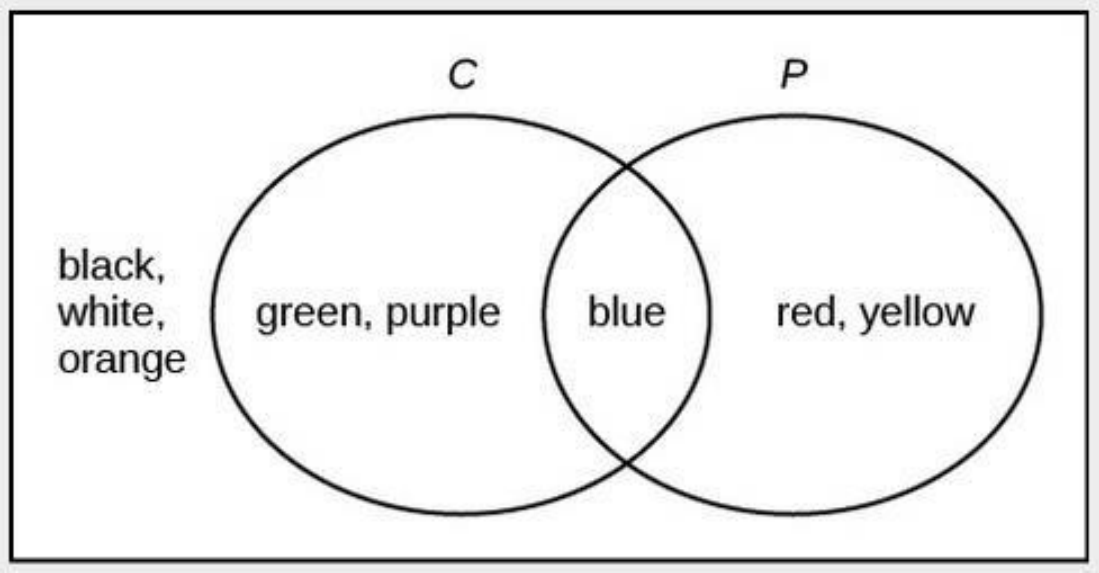
Suppose an experiment has the outcomes 1, 2, 3, ... , 12 where each outcome has an equal chance of occurring. Let event A = {1, 2, 3, 4, 5, 6} and event B = {6, 7, 8, 9}.
Then (A AND B) = {6} and (A OR B) = {1, 2, 3, 4, 5, 6, 7, 8, 9}.
The Venn diagram is as follows:



Venn Diagram

Example

Suppose an experiment has outcomes black, white, red, orange, yellow, green, blue, and purple, where each outcome has an equal chance of occurring. Let event C = {green, blue, purple} and event P = {red, yellow, blue}. Then C AND P = {blue} and C OR P = {green, blue, purple, red, yellow}. Draw a Venn diagram representing this situation.



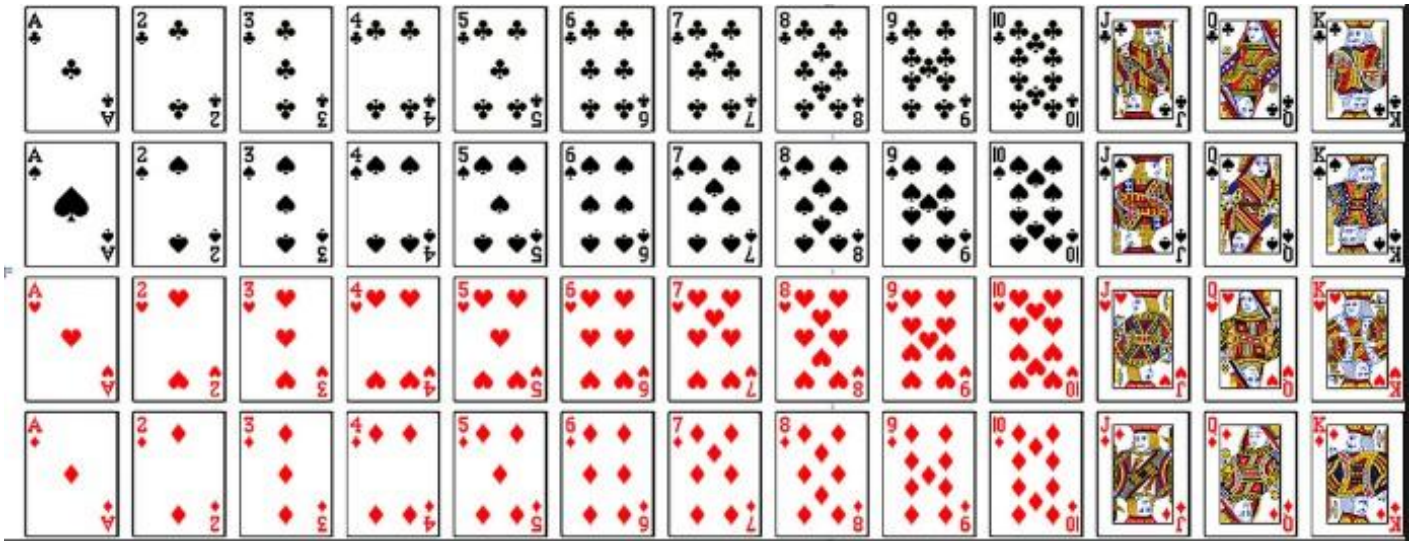
What is probability of getting 5 different cards (no 2, 3, or 4 cards of same value) from a standard deck of cards?

A. 0.7053

B. 0.5040

C. 0.5071

D. 0.08



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$$p(5\,different\ cards) = 1 \cdot \frac{48}{51} \cdot \frac{44}{50} \cdot \frac{40}{49} \cdot \frac{36}{48} = \frac{2112}{4165}$$

$$p(first\ be\ any\ card) = \frac{52}{52}$$

$$p(2nd\ card\ be\ different\ from\ the\ first\ card) = \frac{48}{51}$$

$$p(3rd\ card\ be\ different\ from\ the\ first\ and\ 2nd\ cards) = \frac{44}{50}$$

$$\vdots$$

What is the probability of 10 people in a room, nobody shares a birthday?

A. 0.883

B. 0.535

C. 0.494

D. 0.083

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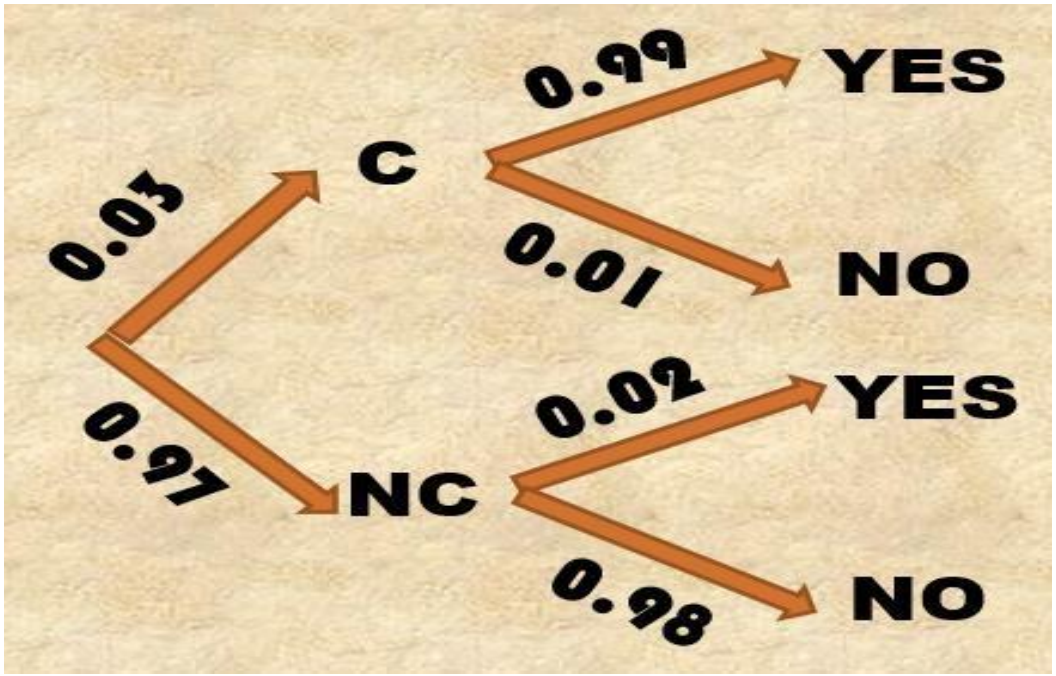
$$p(\textit{nobody shares a birthday}) = \frac{365 \cdot 364 \cdot 363 \cdot 362 \cdot \cdots \cdot 356}{(365)^{10}} = 0.883$$

Bayes' theorem

$$p(A \cap B) = p(A)\,p(B \mid A) \rightarrow p(B \mid A) = \frac{p(A \cap B)}{p(A)}$$

$$\rightarrow p(B \mid A) = \frac{p(B)\,p(A \mid B)}{p(A)}$$

Probability to have a type of cancer in a population is 0.03. A test for the disease is 99% accurate if the person has the disease; and it is 98% accurate if the person does not have the disease. What is the probability of a person having the disease if the test result is positive (test result indicates the person has the disease)?



$$P(C) = 0.03$$

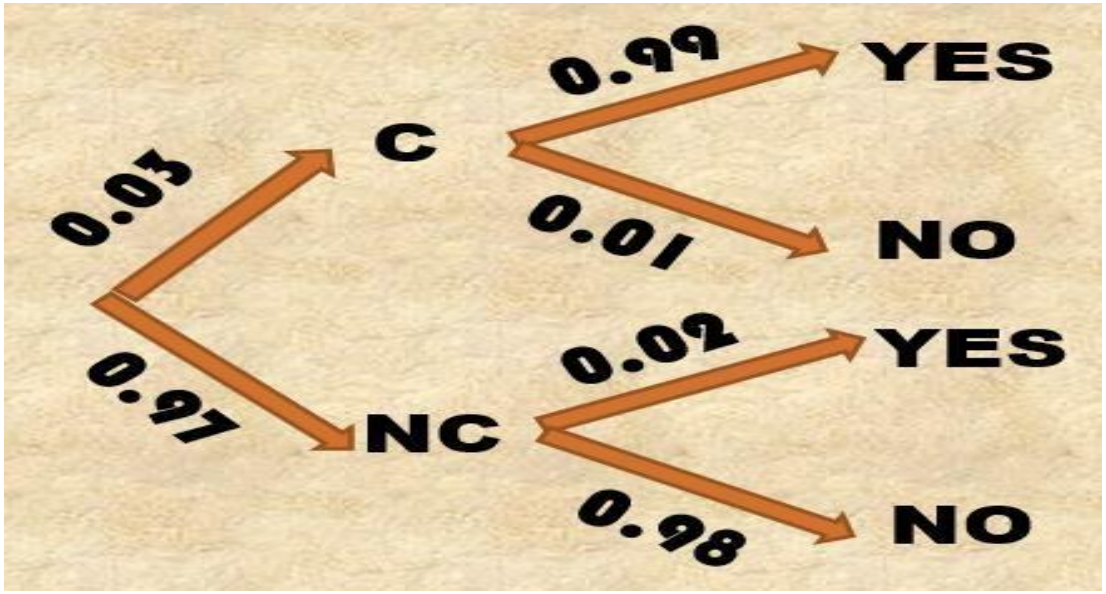
$$P(YES \mid C) = 0.99$$

$$P(NO \mid C) = 0.01$$

$$P(NC) = 0.97$$

$$P(YES \mid NC) = 0.02$$

$$P(NO \mid NC) = 0.98$$



$$p(B \mid A) = \frac{p(B)\,p(A \mid B)}{p(A)}$$

$$p(C \mid YES) = \frac{p(C)\,p(YES \mid C)}{p(YES)} = \frac{(0.03)(0.99)}{p(YES)}$$

$$= \frac{(0.03)(0.99)}{p(YES \mid C)\,p(C) + p(YES \mid NC)\,p(NC)}$$

$$= \frac{(0.03)(0.99)}{(0.99)(0.03) + (0.02)(0.97)} = 0.6049$$

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